Physics Worksheet

Derrivatives and Their Possible use in Physics

Name: _____ **Student #:** _____ **Student Email #:** _____

1. **Quotient Rule**
   Compute: $\frac{d}{dx}\left(\frac{\cos x}{\sin x}\right)$

2. **Chain Rule**
   Compute: $\frac{d}{dx}\ \ln(\cos(x))$

3. **Product Rule + Chain Rule**
   Compute: $\frac{d}{dx}(\cos(x)\sin(x) - \frac{1}{2}\sin(2x))$

4. **Chain Rule + Product Rule**
   Compute: $\frac{d}{dx}(e^{2x\cos(x)})$